# Derivation of the Exponential Moving Average

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## 1 Overview

The essence of the *exponential moving average* (EMA) is to provide a *localized* estimate of a time series as it proceeds. It does this by using a weighted average of the values in a "window" around each point of the series in a way that each summand in the average is multiplied by powers of a factor, $\lambda$, in the interval $(0, 1)$. In what follows we will assume a 1-based index, and that the data series $\mathbf{x}$ has length $N$.

Specifically, the exponential moving average (over a window of length $m$) is defined as:

$$e_n = \sum_{k=0}^{m-1} x_{n-k} w_{k+1} \tag{1}$$

To get the *exponential decay* in weights we assume that $w_{k+1} = \lambda w_k$.[1] Since the parameter $\lambda$ is in the interval $(0, 1)$, it represents a decay factor. Notice that the indices of $\mathbf{x}$ and that of $\mathbf{w}$ go in the "opposite direction". This is done so that the largest weights are associated with the most recent data in the averaging window.

Before we go further, there is a problem with the formula as stated. If $n = 1$; or, in fact, any value less than $m$, the expression $x_{n-k}$ does not make sense for $k \geq n$. (At $n = m$ the window is exactly full: its oldest term is $x_1$.) That is, at the start of the computation and for a little while, there is not a full window of values over a window of length $m$. As there is no information before $x_1$, we interpret these values to be $x_1$. We will apply this principle more generally below. That is, if $\mathbf{z}$ is a data series, we extend the definition of $\mathbf{z}$ so that it is defined for indices less than 1 as:

$$z_k \equiv z_{\max(k,\,1)} \tag{2}$$

While we can use (1) to compute the EMA it is *inefficient*. The purpose of this paper is to derive a recursive formula that is fast to compute. The formula takes advantage of the fact that when the weights are exponentials (powers of $\lambda$) we don't have to recompute the bulk of the sums in the averaging window.

[1] The reason this is called exponential is that it is the discrete analog of continuous exponential decay. To see this let $f(t) = e^{-\beta t}$ for $\beta > 0$. Then sampling this function at regular intervals of say 1 we get the sequence: $\{e^{-\beta}, e^{-2\beta}, e^{-3\beta}, \ldots\}$. Here we can see that the elements of the sequence are powers of a value, $\lambda = e^{-\beta}$. That is, the sequence is nothing other than: $\{\lambda, \lambda^2, \lambda^3, \ldots\}$.

Regardless of the averaging scheme, we want our weights to sum to one.[2] But we also want the weights to be exponential; that is, $w_{k+1} = \lambda w_k$.

Ideally, all we want to do is give the decay factor, $\lambda$. Given $\lambda$, the weights are constructed as follows.

For $i \in [1, m]$ define

$$\hat{w}_i = \lambda^i \tag{3}$$

The $\hat{\mathbf{w}}$ have the property that they decay with $\lambda$; that is, $\lambda \hat{w}_k = \hat{w}_{k+1}$. However, how do we ensure that these values sum to 1 over the window?

To do this, we simply normalize the weights:

$$w_i = \frac{\hat{w}_i}{\sum_{k=1}^{m} \hat{w}_k} \quad \forall i \in [1, m] \tag{4}$$

Consequently,

$$\sum_{i=1}^{m} w_i = 1 \tag{5}$$

In doing this, did the proposed weights, $\mathbf{w}$, lose the decay factor property? The answer is no, the property remains:

$$\lambda w_i = w_{i+1} \quad \forall i \in [1, m-1]. \tag{6}$$

More generally, we can find *all* weights that have a decay factor of $\lambda$ as they are solutions of the following difference equation:

$$w_{n+1} = \lambda\, w_n \tag{7}$$
$$w_1 = \alpha \tag{8}$$

The solution to this difference equation is:

$$w_n = \alpha\, \lambda^{n-1} \quad n \geq 1 \tag{9}$$

However, we need the weights to be *normalized* over a window of length, $m$:

$$\sum_{i=1}^{m} w_i = 1 \tag{10}$$

[2] If we consider the underlying data to be generated from a series of identically distributed random variables; then, treating the series elements as random variables we can view the moving average as an estimate of their mean, $\mu$. To be a good estimate we would like it to be *unbiased*: $\mathbb{E}\left[\sum_{k=0}^{m-1} x_{n-k} w_{k+1}\right] = \mu$. By linearity of the expectation, $\mathbb{E}\left[\sum_{k=0}^{m-1} x_{n-k} w_{k+1}\right] = \sum_{k=0}^{m-1} \mathbb{E}\left[x_{n-k}\right] w_{k+1} = \mu \sum_{k=1}^{m} w_k$; so the estimate is unbiased exactly when the weights sum to 1.

Plugging in our general formula for an exponential sequence we have:

$$\sum_{i=1}^{m} \alpha\lambda^{i-1} = 1 \tag{11}$$

$$\alpha\frac{1-\lambda^m}{1-\lambda} = 1 \tag{12}$$

Therefore, a decay factor, $\lambda$, and averaging window length, $m$, determine *the* initial weight, $\alpha$, and thus determine *the* associated normalized weights:

$$\alpha = \frac{1-\lambda}{1-\lambda^m} \tag{13}$$

**Note.** We will use (2), (5), (6), and (13) in our derivations below.

# 2 Recursive Formula for EMA

We state again the definition of the exponential moving average of $x$ as:

$$e_n = \sum_{k=0}^{m-1} x_{n-k} w_{k+1} \tag{14}$$

In order to find a recursive formula, we examine the next element in the smoothing sequence and try to relate it to the present element at index $n$.

$$
\begin{aligned}
e_{n+1} &= \sum_{k=0}^{m-1} x_{n+1-k} w_{k+1} \\
&= \sum_{k=0}^{m-1} x_{n-(k-1)} w_{k+1} \\
&= \sum_{k=-1}^{m-2} x_{n-k} w_{k+2} \\
&= \sum_{k=0}^{m-2} x_{n-k} w_{k+2} + x_{n+1} w_1 \\
&= \lambda \sum_{k=0}^{m-2} x_{n-k} w_{k+1} + x_{n+1} w_1 \quad (\text{Since } \lambda w_{k+1} = w_{k+2}) \\
&= \lambda \sum_{k=0}^{m-1} x_{n-k} w_{k+1} + x_{n+1} w_1 - \lambda x_{n-(m-1)} w_m \\
&= \lambda \sum_{k=0}^{m-1} x_{n-k} w_{k+1} + x_{n+1} w_1 - \lambda x_{(n+1)-m} w_m
\end{aligned}
$$

Finally, we may write[3]

$$
e_{n+1} = \lambda \left( e_n - x_{(n+1)-m} w_m \right) + \alpha x_{n+1} \quad \forall n \in [1, N-1] \tag{15}
$$

$$
e_1 = x_1 \tag{16}
$$

The corresponding 0-based index formula is:[4]

$$
e_n = \lambda \left( e_{n-1} - x_{n-m} w_{m-1} \right) + \alpha x_n \quad \forall n \in [1, N-1] \tag{17}
$$

$$
e_0 = x_0 \tag{18}
$$

# 3 Recursive Formula for a Local Standard Deviation

In practice, one also wants to know a localized standard deviation of the process. In applications, one can provide "Bollinger" bands around the EMA estimate. These bands are (typically) some multiple of the standard deviation of the process.

[3] From the definition of $e_n$ and using (2), (5), and (13), $e_1 = \sum_{k=0}^{m-1} x_{1-k} w_{k+1} = x_1 \sum_{k=0}^{m-1} w_{k+1} = x_1 \sum_{k=1}^{m} w_k = x_1$.

[4] We write the formulas down for both 1- and 0-based vector indices because computer languages typically use one or the other. In the 0-based formulas the weights are re-indexed as well: they are $w_0, \ldots, w_{m-1}$, so the first (largest) weight is $\alpha = w_0$ and the oldest weight is $w_{m-1}$.

We derive a localized standard deviation of the process starting with the definition of the exponential moving average and its moving variance.[5]

$$e_n = \sum_{k=0}^{m-1} x_{n-k} w_{k+1} \tag{19}$$

$$v_n = \sum_{k=0}^{m-1} (x_{n-k} - e_{n-k})^2 w_{k+1} \tag{20}$$

Note that the variance (20) measures squared deviation of each $x_{n-k}$ from its own time's EMA $e_{n-k}$, rather than from the current $e_n$. This choice lets the recursive update for $v_n$ be derived using the same machinery as $e_n$.[6]

Note, that the form of the two equations is identical; that is, they both have the form:

$$g_n = \sum_{k=0}^{m-1} f(x_{n-k} - c_{n-k}) w_{k+1} \tag{21}$$

In equation (19), $c_{n-k} \equiv 0$; while in equation (20), $c_{n-k} = e_{n-k}$.

One can find a recursive formula $g_n$ in *exactly* the same way as was done for (19). This is done by reworking the derivation of the recursive formula for (19), replacing $x_{n-k}$ with $f(x_{n-k} - c_{n-k})$.

The recursive formula for $g_n$ is

$$g_{n+1} = \lambda \left( g_n - f(x_{(n+1)-m} - c_{(n+1)-m}) w_m \right) + \alpha f(x_{n+1} - c_{n+1}) \quad \forall n \in [1, N-1] \tag{22}$$

$$g_1 = f(x_1 - c_1) \tag{23}$$

The 0-based index formula is:

$$g_n = \lambda \left( g_{n-1} - f(x_{n-m} - c_{n-m}) w_{m-1} \right) + \alpha f(x_n - c_n) \quad \forall n \in [1, N-1] \tag{24}$$

$$g_0 = f(x_0 - c_0) \tag{25}$$

The corresponding $f$ for (20) is $f(x) = x^2$; consequently, its recursive formula is

$$v_{n+1} = \lambda \left( v_n - (x_{(n+1)-m} - e_{(n+1)-m})^2 w_m \right) + \alpha (x_{n+1} - e_{n+1})^2 \quad \forall n \in [1, N-1] \tag{26}$$

$$v_1 = (x_1 - e_1)^2 = 0 \tag{27}$$

[5] This variance formula needs to be corrected by a factor which we leave to the end. The correction is a factor that is used to make the estimated standard deviation of the moving average *unbiased*. The formula for this factor is independent of the structure of the weights. This means that it can be pre-computed independently of the data series.

[6] The unbiasing factor $1 - \sum_{i=1}^{m} w_i^2$ derived in the appendix assumes a single common center, $\bar{x} = \sum_i x_i w_i$, applied to every summand. Here each summand uses its own center $e_{n-k}$, so the correction is applied as a working approximation; an exact unbiased correction for centers $e_{n-k}$ would require a separate derivation.

The corresponding 0-based index formula is:

$$v_n = \lambda\left(v_{n-1} - (x_{n-m} - e_{n-m})^2 w_{m-1}\right) + \alpha(x_n - e_n)^2 \quad \forall n \in [1, N-1] \tag{28}$$

$$v_0 = (x_0 - e_0)^2 = 0 \tag{29}$$

Finally, the localized estimate of the standard deviation of the *process* is:[7]

$$s_n = \sqrt{\frac{v_n}{1 - \sum_{i=1}^{m} w_i^2}} \tag{30}$$

The variance corrective factor[8], $1 - \sum_{i=1}^{m} w_i^2$, is a one-time calculation independent of $n$. This factor is the standard correction when determining empirical variance of a weighted average.

**Note.** In practice, one may wish to use some prior estimate of the standard deviation of the process and smoothly move from that to the estimate above over the averaging windows when $n \in [1, m]$. For the same reason – $v_1 = 0$ and a partially filled window – the early values $s_n$, for $n$ up to about $m$, should not be relied upon on their own.

# 4 How to Specify the $\lambda$ Factor

When using the EMA, practitioners do not specify the factor $\lambda$ directly; instead they provide the "half-life", $H$, of the decay. The *half-life* is the index $H$ such that $w_H/w_1 = 1/2$; that is, the weight halves in going from index 1 to index $H$.[9] Substituting $w_n = \alpha\lambda^{n-1}$ gives:

$$\frac{w_H}{w_1} = \frac{\alpha\lambda^{H-1}}{\alpha} = \lambda^{H-1} = \frac{1}{2} \tag{31}$$

For a given $\lambda$, an integer $H$ satisfying this exactly may not exist; in that direction one takes the closest integer.[10]

In our case we are interested in the reverse; namely, given a half-life, $H$, determine the decay factor $\lambda$. To do this we solve the equation $\lambda^{H-1} = \frac{1}{2}$ for $\lambda$. Taking the log of this equation we have:[11]

[7] The window length, $m$, must be greater than 1. Note that this is an estimate of the local standard deviation of the underlying process – which is what Bollinger bands require – not the standard deviation of the estimator $e_n$ itself; the latter is the smaller quantity $\sigma\sqrt{\sum_{i=1}^{m} w_i^2}$.

[8] This factor is derived in the appendix.

[9] This convention measures the halving over $H-1$ steps. Some libraries (e.g., the `ewm` functions of pandas) instead define the half-life by $\lambda^H = 1/2$ – halving over $H$ steps – which yields $\lambda = 2^{-1/H}$. The two conventions differ by one step: for $H = 5$, ours gives $\lambda \approx 0.8409$ while the other gives $\lambda = 2^{-1/5} \approx 0.8706$.

[10] Note that $H$ must be greater than 1 and no more than the window length, $m$.

[11] The log function used below is the logarithm base $e$.

$$
\begin{aligned}
\log(\lambda^{H-1}) &= \log\left(\frac{1}{2}\right) \\
(H-1)\log(\lambda) &= \log(1) - \log(2) \\
\log(\lambda) &= -\frac{\log(2)}{H-1}
\end{aligned}
$$

The formula to determine $\lambda$ given the half-life $H$ is:

$$\lambda = e^{-\frac{\log(2)}{H-1}} \tag{32}$$

For example, if the window of the moving average is 10, you might decide that the half-life occurs around the middle of the window. Setting $H$ to be 5, the decay factor, $\lambda$ is $e^{-\frac{\log(2)}{4}} \approx 0.8409$.

# A Unbiased Estimator for a Sequence of IID Random Variables

In this section we derive the corrective formula for computing the empirical variance.

In what follows we assume that we are given a sequence of independent identically distributed random variables (IID): $\{x_i\}_{i=1}^N$ and corresponding weights: $\{w_i\}_{i=1}^N$. We assume that the random variables have a mean and a variance:

$$\mathbb{E}[x_i] = \mu \quad \forall i \in [1, N] \tag{33}$$

$$\mathbb{E}\left[(x_i - \mu)^2\right] = \sigma^2 \quad \forall i \in [1, N] \tag{34}$$

We assume that the weights are non-negative and are normalized:[12]

- $\sum_{i=1}^N w_i = 1$
- $\forall i, 0 \leq w_i < 1$

If we take the expectation of the weighted sum, $\sum_{i=1}^N x_i w_i$ we find that it is $\mu$:[13]

$$\mathbb{E}\left[\sum_{i=1}^N x_i w_i\right] = \sum_{i=1}^N \mathbb{E}[x_i w_i] = \sum_{i=1}^N \mathbb{E}[x_i] w_i = \mu \sum_{i=1}^N w_i = \mu \tag{35}$$

The weighted sum represents an estimate of the true mean of the IID random variables, which is $\mu$. And in this case we see that it is a good estimate in that the expected value of the weighted combination of the random variables is $\mu$.

We refer to such "good" estimates as *unbiased estimators*.

Now consider the sum $\sum_{i=1}^N (x_i - \mu)^2 w_i$. People use this as an estimate of the variance of the random variables. Is this a "good" (unbiased) estimate? Yes, it is as: $\mathbb{E}\left[\sum_{i=1}^N (x_i - \mu)^2 w_i\right] = \sum_{i=1}^N \mathbb{E}\left[(x_i - \mu)^2\right] w_i = \sigma^2 \sum_{i=1}^N w_i = \sigma^2$.

But what if we replace the true mean, $\mu$, above with our estimate: $\bar{x} = \sum_{i=1}^N x_i w_i$? In this case it turns out that the resulting estimate for the variance is *biased*. However, it happens that we are "off" by a certain factor and that factor depends only on the weights we use.[14]

To find this factor we examine the expected value of the straightforward formula one would imagine using to estimate the variance of the random variables.

[12] We also assume that the weight is not concentrated at a single index.

[13] This is due to the linearity of the expectation operator: $\mathbb{E}[ax + by + c] = a\mathbb{E}[x] + b\mathbb{E}[y] + c$, provided $a, b, c$ are constants.

[14] In fact, the factor doesn't depend on the ordering of the weights.

$$
\begin{aligned}
\mathbb{E}\left[\sum_{i=1}^{N}(x_i-\bar{x})^2 w_i\right] &= \sum_{i=1}^{N}\mathbb{E}\left[(x_i-\bar{x})^2\right] w_i \\
&= \sum_{i=1}^{N}\mathbb{E}\left[\left((x_i-\mu)+(\mu-\bar{x})\right)^2\right] w_i \\
&= \sum_{i=1}^{N}\mathbb{E}\left[(x_i-\mu)^2-2(x_i-\mu)(\bar{x}-\mu)+(\bar{x}-\mu)^2\right] w_i \\
&= \sum_{i=1}^{N}\left(\mathbb{E}\left[(x_i-\mu)^2\right]+\mathbb{E}\left[(\bar{x}-\mu)^2\right]-2\mathbb{E}\left[(x_i-\mu)(\bar{x}-\mu)\right]\right) w_i
\end{aligned} \tag{36}
$$

We separately compute the expectation of each of the three summands.

$$
\mathbb{E}\left[(x_i-\mu)^2\right] = \sigma^2 \tag{37}
$$

$$
\begin{aligned}
\mathbb{E}\left[(\bar{x}-\mu)^2\right] &= \mathbb{E}\left[\left(\left(\sum_{j=1}^{N} x_j w_j\right) - \mu\right)^2\right] \\
&= \mathbb{E}\left[\left(\sum_{j=1}^{N} (x_j w_j - \mu w_j)\right)^2\right] \\
&= \mathbb{E}\left[\left(\sum_{j=1}^{N} (x_j - \mu) w_j\right)^2\right] \\
&= \mathbb{E}\left[\left(\sum_{j=1}^{N} (x_j - \mu) w_j\right)\left(\sum_{k=1}^{N} (x_k - \mu) w_k\right)\right] \\
&= \mathbb{E}\left[\sum_{j=1}^{N}\sum_{k=1}^{N} (x_j - \mu)(x_k - \mu) w_j w_k\right] \\
&= \left(\sum_{j=1}^{N}\sum_{k=1}^{N} \mathbb{E}\left[(x_j - \mu)(x_k - \mu)\right] w_j w_k\right) \\
&= \left(\sum_{j=1}^{N}\sum_{k=1}^{N} \sigma^2 w_j w_k \delta_{j,k}\right) \\
&= \left(\sum_{j=1}^{N} \sigma^2 w_j^2\right) \\
&= \sigma^2 \left(\sum_{j=1}^{N} w_j^2\right)
\end{aligned} \tag{38}
$$

$$
\begin{aligned}
\mathbb{E}\left[(x_i-\mu)(\bar{x}-\mu)\right] &= \mathbb{E}\left[(x_i-\mu)\left(\left(\sum_{j=1}^{N} x_j w_j\right)-\mu\right)\right] \\
&= \mathbb{E}\left[(x_i-\mu)\left(\sum_{j=1}^{N}(x_j-\mu)w_j\right)\right] \\
&= \mathbb{E}\left[\sum_{j=1}^{N}(x_i-\mu)(x_j-\mu)w_j\right] \\
&= \sum_{j=1}^{N}\mathbb{E}\left[(x_i-\mu)(x_j-\mu)\right]w_j \\
&= \sum_{j=1}^{N}\sigma^2 w_j \delta_{i,j} \\
&= \sigma^2 w_i
\end{aligned} \tag{39}
$$

Substituting these three results into (37):

$$
\begin{aligned}
\mathbb{E}\left[\sum_{i=1}^{N}(x_i-\bar{x})^2 w_i\right] &= \sum_{i=1}^{N}\left(\sigma^2+\sigma^2\sum_{j=1}^{N}w_j^2-2\sigma^2 w_i\right)w_i \\
&= \sigma^2+\sigma^2\sum_{j=1}^{N}w_j^2-2\sigma^2\sum_{i=1}^{N}w_i^2 \\
&= \sigma^2\left(1-\sum_{i=1}^{N}w_i^2\right)
\end{aligned}
$$

Therefore, to make $\sum_{i=1}^{N}(x_i-\bar{x})^2 w_i$ an unbiased estimator of the variance $\sigma^2$ we need to correct by dividing by the factor $\left(1-\sum_{i=1}^{N}w_i^2\right)$.[15]

Consequently, $\frac{1}{1-\sum_{i=1}^{N}w_i^2}\sum_{i=1}^{N}(x_i-\bar{x})^2 w_i$ is an unbiased estimator for the variance of any of the IID variables: $\{x_i\}_{i=1}^{N}$. The unbiased standard deviation is then

$$
\sqrt{\frac{1}{1-\sum_{i=1}^{N}w_i^2}\sum_{i=1}^{N}(x_i-\bar{x})^2 w_i} \tag{40}
$$

[15]Note, this factor is non-zero when the weights are not concentrated at a single index. As a consequence, the formula requires $N>1$.

What is the formula when the weights are evenly distributed? That would mean that the weights are $w_i = \frac{1}{N}$. In this case the estimate for the standard deviation of the random variables is:

$$
\begin{aligned}
\sqrt{\frac{1}{1-\sum_{i=1}^{N}\left(\frac{1}{N}\right)^2}\sum_{i=1}^{N}(x_i-\bar{x})^2\frac{1}{N}} &= \sqrt{\frac{1}{1-\frac{1}{N}}\frac{1}{N}\sum_{i=1}^{N}(x_i-\bar{x})^2} \\
&= \sqrt{\frac{1}{\frac{N-1}{N}}\frac{1}{N}\sum_{i=1}^{N}(x_i-\bar{x})^2} \\
&= \sqrt{\frac{N}{N-1}\frac{1}{N}\sum_{i=1}^{N}(x_i-\bar{x})^2} \\
&= \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(x_i-\bar{x})^2}
\end{aligned}
$$

We see that we end up with the usual "sample" standard deviation.

## A.1 Bias of Arbitrary Sample Moments.

We now proceed in the same way as above to compute the bias of more general moments. Throughout this subsection we assume the central moments $M_k \equiv \mathbb{E}\left[(x_i-\mu)^k\right]$ exist (are finite) for all $k \leq r$. The strategy mirrors the second-moment derivation: rewrite $x_i - \bar{x} = (x_i - \mu) + (\mu - \bar{x})$ and expand the $r^{th}$ power via the binomial theorem. The resulting power of $\sum_l (x_l - \mu) w_l$ is then expanded by the multinomial theorem, splitting off the $l = i$ term so that independence may be used when the expectation is distributed across the product.

$$
\begin{aligned}
\mathbb{E}\left[\sum_{i=1}^{N}(x_i-\bar{x})^r w_i\right] &= \sum_{i=1}^{N}\mathbb{E}\left[(x_i-\bar{x})^r\right]w_i \qquad (41)\\
&= \sum_{i=1}^{N}\mathbb{E}\left[((x_i-\mu)+(\mu-\bar{x}))^r\right]w_i\\
&= \sum_{i=1}^{N}\mathbb{E}\left[\sum_{j=0}^{r}(-1)^j\binom{r}{j}(x_i-\mu)^{r-j}(\bar{x}-\mu)^j\right]w_i\\
&= \sum_{i=1}^{N}\sum_{j=0}^{r}(-1)^j\binom{r}{j}\mathbb{E}\left[(x_i-\mu)^{r-j}(\bar{x}-\mu)^j\right]w_i\\
&= \sum_{i=1}^{N}\sum_{j=0}^{r}(-1)^j\binom{r}{j}\mathbb{E}\left[(x_i-\mu)^{r-j}\left(\sum_{l=1}^{N}(x_l-\mu)w_l\right)^j\right]w_i\\
&= \sum_{i=1}^{N}\sum_{j=0}^{r}(-1)^j\binom{r}{j}\mathbb{E}\left[(x_i-\mu)^{r-j}\left((x_i-\mu)w_i+\sum_{\substack{l\in[1,N]\\ l\neq i}}(x_l-\mu)w_l\right)^j\right]w_i\\
&= \sum_{i=1}^{N}\sum_{j=0}^{r}(-1)^j\binom{r}{j}\mathbb{E}\left[(x_i-\mu)^{r-j}\sum_{k=0}^{j}\binom{j}{k}(x_i-\mu)^{j-k}w_i^{j-k}\left(\sum_{\substack{l\in[1,N]\\ l\neq i}}(x_l-\mu)w_l\right)^k\right]w_i\\
&= \sum_{i=1}^{N}\sum_{j=0}^{r}\sum_{k=0}^{j}(-1)^j\binom{r}{j}\binom{j}{k}\mathbb{E}\left[(x_i-\mu)^{r-k}w_i^{j-k}\left(\sum_{\substack{l\in[1,N]\\ l\neq i}}(x_l-\mu)w_l\right)^k\right]w_i\\
&= \sum_{i=1}^{N}\sum_{j=0}^{r}\sum_{k=0}^{j}(-1)^j\binom{r}{j}\binom{j}{k}\mathbb{E}\left[(x_i-\mu)^{r-k}\right]\mathbb{E}\left[\left(\sum_{\substack{l\in[1,N]\\ l\neq i}}(x_l-\mu)w_l\right)^k\right]w_i^{j+1-k}\\
&= \sum_{i=1}^{N}\sum_{j=0}^{r}(-1)^j\binom{r}{j}\mathbb{E}\left[(x_i-\mu)^r\right]w_i^{j+1}\\
&\quad+\sum_{i=1}^{N}\sum_{j=2}^{r}\sum_{k=2}^{j}(-1)^j\binom{r}{j}\binom{j}{k}\mathbb{E}\left[(x_i-\mu)^{r-k}\right]\mathbb{E}\left[\left(\sum_{\substack{l\in[1,N]\\ l\neq i}}(x_l-\mu)w_l\right)^k\right]w_i^{j+1-k}
\end{aligned}
$$

Separating out the $j=0$ term (which yields the bare $M_r$) and noting the $k=1$ contributions vanish because

$\mathbb{E}\left[x_l - \mu\right] = 0$, we restrict the inner multinomial sum to indices with $k_n \neq 1$ and obtain

$$\begin{aligned}
\mathbb{E}\left[\sum_{i=1}^{N}(x_i-\bar{x})^r w_i\right] &= M_r + M_r\sum_{j=1}^{r}(-1)^j\binom{r}{j}\sum_{i=1}^{N}w_i^{j+1} + \sum_{i=1}^{N}\sum_{j=2}^{r}\sum_{k=2}^{j}(-1)^j\binom{r}{j}\binom{j}{k}M_{r-k} \\
&\quad\times\mathbb{E}\left[\sum_{\substack{k_1+k_2+\cdots+k_N=k\\ \forall n,k_n\geq 0\\ k_i=0}}\binom{k}{k_1,k_2,\,\ldots\,,k_N}\prod_{\substack{l\in[1,N]\\ l\neq i}}(x_l-\mu)^{k_l}w_l^{k_l}\right]w_i^{j+1-k} \\
&= M_r + M_r\sum_{j=1}^{r}(-1)^j\binom{r}{j}\sum_{i=1}^{N}w_i^{j+1} + \sum_{i=1}^{N}\sum_{j=2}^{r}\sum_{k=2}^{j}(-1)^j\binom{r}{j}\binom{j}{k}M_{r-k} \\
&\quad\times\sum_{\substack{k_1+k_2+\cdots+k_N=k\\ \forall n,k_n\geq 0\,\wedge\,k_n\neq 1\\ k_i=0}}\binom{k}{k_1,k_2,\,\ldots\,,k_N}\left(\prod_{\substack{l\in[1,N]\\ l\neq i}}\mathbb{E}\left[(x_l-\mu)^{k_l}\right]w_l^{k_l}\right)w_i^{j+1-k} \\
&= M_r + M_r\sum_{j=1}^{r}(-1)^j\binom{r}{j}\sum_{i=1}^{N}w_i^{j+1} + \sum_{i=1}^{N}\sum_{j=2}^{r}\sum_{k=2}^{j}(-1)^j\binom{r}{j}\binom{j}{k}M_{r-k} \\
&\quad\times\sum_{\substack{k_1+k_2+\cdots+k_N=k\\ \forall n,k_n\geq 0\,\wedge\,k_n\neq 1\\ k_i=0}}\binom{k}{k_1,k_2,\,\ldots\,,k_N}\left(\prod_{l=1}^{N}M_{k_l}w_l^{k_l}\right)w_i^{j+1-k}
\end{aligned} \tag{42}$$

Here, we use the notation $M_r \equiv \mathbb{E}\left[(x_i-\mu)^r\right]$ for the $r^{th}$ central moment of the distribution of the $\{x_i\}_{i=1}^N$ (so that $M_2 = \sigma^2$).

We now drop the $k_i = 0$ constraint by allowing $k_i$ to take any value in $\{0\}\cup\{2,3,\ldots,k\}$ – this makes the *first* inner sum below independent of $i$ – and subtract back the $k_i \in \{2,\ldots,k\}$ contributions as a separate $z$-indexed sum, with $z = k_i$. In the subtracted term, the remaining $N-1$ slots $k_1,\ldots,k_{N-1}$ correspond to the indices $l \neq i$, relabeled as $1,\ldots,N-1$; that term therefore still depends on $i$, both through this exclusion and through the factor $w_i^{j+z+1-k}$. The multinomial coefficient with one slot equal to $z$ factors as

$\binom{k}{z}\binom{k-z}{k_1,\ldots,k_{N-1}}$, which produces the $\binom{k}{z}$ prefactor in the subtraction below.[16]

$$\begin{aligned}\mathbb{E}\left[\sum_{i=1}^{N}(x_i-\bar{x})^r w_i\right] &= M_r + M_r\sum_{j=1}^{r}(-1)^j\binom{r}{j}\sum_{i=1}^{N}w_i^{j+1} + \sum_{j=2}^{r}\sum_{k=2}^{j}(-1)^j\binom{r}{j}\binom{j}{k}M_{r-k} \\ &\times\left[\sum_{i=1}^{N}w_i^{j+1-k}\left(\sum_{\substack{k_1+k_2+\cdots+k_N=k\\ \forall n, k_n\geq 0\,\wedge\, k_n\neq 1}}\binom{k}{k_1,k_2,\ldots,k_N}\prod_{l=1}^{N}M_{k_l}w_l^{k_l}\right)\right. \\ &\left.-\sum_{z=2}^{k}\binom{k}{z}M_z\left(\sum_{i=1}^{N}w_i^{j+z+1-k}\sum_{\substack{k_1+k_2+\cdots+k_{N-1}=k-z\\ \forall n, k_n\geq 0\,\wedge\, k_n\neq 1}}\binom{k-z}{k_1,k_2,\ldots,k_{N-1}}\prod_{l=1}^{N-1}M_{k_l}w_l^{k_l}\right)\right]\end{aligned} \tag{43}$$

We check the case $r=2$ to see if it coincides with the previous section:

$$\begin{aligned}\mathbb{E}\left[\sum_{i=1}^{N}(x_i-\bar{x})^2 w_i\right] &= \sigma^2 + \sigma^2\sum_{j=1}^{2}(-1)^j\binom{2}{j}\sum_{i=1}^{N}w_i^{j+1} + \sum_{j=2}^{2}\sum_{k=2}^{j}(-1)^j\binom{2}{j}\binom{j}{k}M_{2-k} \\ &\times\sum_{i=1}^{N}\left[\sum_{\substack{k_1+k_2+\cdots+k_N=k\\ \forall n, k_n\geq 0\,\wedge\, k_n\neq 1}}\binom{k}{k_1,k_2,\ldots,k_N}\left(\prod_{l=1}^{N}M_{k_l}w_l^{k_l}\right)w_i^{j+1-k}\right. \\ &\left.-\sum_{z=2}^{k}\binom{k}{z}M_z\sum_{\substack{k_1+k_2+\cdots+k_{N-1}=k-z\\ \forall n, k_n\geq 0\,\wedge\, k_n\neq 1}}\binom{k-z}{k_1,k_2,\ldots,k_{N-1}}\left(\prod_{l=1}^{N-1}M_{k_l}w_l^{k_l}\right)w_i^{j+z+1-k}\right]\end{aligned} \tag{44}$$

This becomes

$$\begin{aligned}\mathbb{E}\left[\sum_{i=1}^{N}(x_i-\bar{x})^2 w_i\right] &= \sigma^2 + \sigma^2\sum_{j=1}^{2}(-1)^j\binom{2}{j}\sum_{i=1}^{N}w_i^{j+1} \\ &+\sum_{\substack{k_1+k_2+\cdots+k_N=2\\ \forall n, k_n\geq 0\,\wedge\, k_n\neq 1}}\binom{2}{k_1,k_2,\ldots,k_N}\left(\prod_{l=1}^{N}M_{k_l}w_l^{k_l}\right)\left(\sum_{i=1}^{N}w_i\right)-\sigma^2\left(\sum_{i=1}^{N}w_i^3\right)\end{aligned} \tag{45}$$

Which is

$$\mathbb{E}\left[\sum_{i=1}^{N}(x_i-\bar{x})^2 w_i\right] = \sigma^2+\sigma^2\left(-2\sum_{i=1}^{N}w_i^2+\sum_{i=1}^{N}w_i^3\right)+\sigma^2\sum_{l=1}^{N}w_l^2\left(\sum_{i=1}^{N}w_i\right)-\sigma^2\left(\sum_{i=1}^{N}w_i^3\right) \tag{46}$$

[16] We use the standard interpretation of any sum of the form, $\sum_{i=n}^{m} f(i)$, to be 0 if $m<n$.

This becomes

$$\mathbb{E}\left[\sum_{i=1}^{N}(x_i-\bar{x})^2 w_i\right]=\sigma^2\left(1-\sum_{i=1}^{N}w_i^2\right) \tag{47}$$

As a more complicated example we try $r=3$. In this case the formula is:

$$\begin{aligned}\mathbb{E}\left[\sum_{i=1}^{N}(x_i-\bar{x})^3 w_i\right] &= M_3+M_3\sum_{j=1}^{3}(-1)^j\binom{3}{j}\sum_{i=1}^{N}w_i^{j+1}+\sum_{j=2}^{3}\sum_{k=2}^{j}(-1)^j\binom{3}{j}\binom{j}{k}M_{3-k} \\ &\times\sum_{i=1}^{N}\left[\sum_{\substack{k_1+k_2+\cdots+k_N=k\\ \forall n,k_n\geq 0\,\wedge\,k_n\neq 1}}\binom{k}{k_1,k_2,\ldots,k_N}\left(\prod_{l=1}^{N}M_{k_l}w_l^{k_l}\right)w_i^{j+1-k}\right. \\ &\left.-\sum_{z=2}^{k}\binom{k}{z}M_z\sum_{\substack{k_1+k_2+\cdots+k_{N-1}=k-z\\ \forall n,k_n\geq 0\,\wedge\,k_n\neq 1}}\binom{k-z}{k_1,k_2,\ldots,k_{N-1}}\left(\prod_{l=1}^{N-1}M_{k_l}w_l^{k_l}\right)w_i^{j+z+1-k}\right]\end{aligned} \tag{48}$$

Which is

$$\mathbb{E}\left[\sum_{i=1}^{N}(x_i-\bar{x})^3 w_i\right]=M_3+M_3\left(-3\sum_{i=1}^{N}w_i^2+3\sum_{i=1}^{N}w_i^3-\sum_{i=1}^{N}w_i^4\right)-M_3\left(\sum_{l=1}^{N}w_l^3\left(\sum_{i=1}^{N}w_i\right)-\sum_{i=1}^{N}w_i^4\right) \tag{49}$$

Or,

$$\mathbb{E}\left[\sum_{i=1}^{N}(x_i-\bar{x})^3 w_i\right]=M_3\left(1-3\sum_{i=1}^{N}w_i^2+2\sum_{i=1}^{N}w_i^3\right) \tag{50}$$

Therefore, the corresponding unbiased estimator for the $3^{rd}$ central moment is:[17]

$$\frac{\sum_{i=1}^{N}(x_i-\bar{x})^3 w_i}{1-3\sum_{i=1}^{N}w_i^2+2\sum_{i=1}^{N}w_i^3} \tag{51}$$

For $r>3$ the bookkeeping is more involved. We illustrate the case $r=4$ below. Substituting directly into the general formula above gets unwieldy; we instead organize the computation by the value of $k$, the power of $\sum_{l\neq i}(x_l-\mu)w_l$ that appears.

Let $y_i=x_i-\mu$ and $z=\bar{x}-\mu=\sum_{l=1}^{N}y_l w_l$. Split $z=y_i w_i+s_i$, where

$$s_i\equiv\sum_{\substack{l\in[1,N]\\ l\neq i}}y_l w_l$$

[17] As mentioned in the case of the second moment, this formula does not apply when the weights are concentrated at a single index. Consequently, we must have $N>1$.

is independent of $y_i$. Expanding $(x_i - \bar{x})^4 = (y_i - z)^4$ via the binomial theorem, and then $z^k = (y_i w_i + s_i)^k$ via a second application, gives:

$$\mathbb{E}\left[\sum_{i=1}^{N}(x_i - \bar{x})^4 w_i\right] = \sum_{j=0}^{4}\sum_{k=0}^{j}(-1)^j \binom{4}{j}\binom{j}{k} M_{4-k} \sum_{i=1}^{N} w_i^{j-k+1}\mathbb{E}\left[s_i^k\right]. \tag{52}$$

Let $W_j = \sum_{i=1}^N w_i^j$. We collect the contributions by the value of $k$:

- $k = 0$: $\mathbb{E}\left[s_i^0\right] = 1$, giving $M_4 \sum_{j=0}^4 (-1)^j \binom{4}{j} W_{j+1} = M_4(1 - 4W_2 + 6W_3 - 4W_4 + W_5)$.
- $k = 1$: $\mathbb{E}\left[s_i^1\right] = 0$.
- $k = 2$: $\mathbb{E}\left[s_i^2\right] = M_2(W_2 - w_i^2)$, $M_{4-k} = M_2$, summed over $j \in \{2, 3, 4\}$: $M_2^2\big[\,6(W_2 - W_3) - 12(W_2^2 - W_4) + 6(W_2W_3 - W_5)\,\big]$.
- $k = 3$: $M_{4-k} = M_1 = 0$.
- $k = 4$: only $j = 4$ contributes, with $\mathbb{E}\left[s_i^4\right] = M_4(W_4 - w_i^4) + 6M_2^2\big[\,\frac{1}{2}(W_2^2 - W_4) - w_i^2 W_2 + w_i^4\,\big]$, giving $M_4(W_4 - W_5) + 6M_2^2\big[\,\frac{1}{2}(W_2^2 - W_4) - W_2W_3 + W_5\,\big]$.

Adding these (the $W_5$ terms cancel in the $M_4$ piece; the $W_2W_3$ and $W_5$ terms cancel in the $M_2^2$ piece, and $6 \cdot \frac{1}{2}(W_2^2 - W_4) = 3W_2^2 - 3W_4$):

$$\mathbb{E}\left[\sum_{i=1}^{N}(x_i - \bar{x})^4 w_i\right] = M_4(1 - 4W_2 + 6W_3 - 3W_4) + 3M_2^2(2W_2 - 2W_3 - 3W_2^2 + 3W_4). \tag{53}$$

As a check, at $N = 2$ with $w_1 = w_2 = \frac{1}{2}$ ($W_2 = \frac{1}{2}$, $W_3 = \frac{1}{4}$, $W_4 = \frac{1}{8}$): the $M_4$ coefficient is $1 - 2 + \frac{3}{2} - \frac{3}{8} = \frac{1}{8}$, and the $M_2^2$ coefficient is $3(1 - \frac{1}{2} - \frac{3}{4} + \frac{3}{8}) = \frac{3}{8}$, matching the direct evaluation $\frac{M_4}{8} + \frac{3M_2^2}{8}$.

The corresponding relation for the kurtosis, $M_4/M_2^2$, is:

$$\frac{M_4}{M_2^2} = \frac{\mathbb{E}\left[\sum_{i=1}^N (x_i - \bar{x})^4 w_i\right]/M_2^2 - 3(2W_2 - 2W_3 - 3W_2^2 + 3W_4)}{1 - 4W_2 + 6W_3 - 3W_4}. \tag{54}$$

Since we don't know $M_2 = \sigma^2$, we use our unbiased estimate $\widehat{M}_2 = \frac{\sum_{j=1}^N (x_j - \bar{x})^2 w_j}{1 - W_2}$. We also replace $\mathbb{E}\left[\sum_{i=1}^N (x_i - \bar{x})^4 w_i\right]$ with the sample estimate $\sum_{i=1}^N (x_i - \bar{x})^4 w_i$. Our estimate of the kurtosis is then:

$$\frac{M_4}{M_2^2} \approx \frac{\dfrac{(1 - W_2)^2 \sum_{i=1}^N (x_i - \bar{x})^4 w_i}{\left(\sum_{j=1}^N (x_j - \bar{x})^2 w_j\right)^2} - 3(2W_2 - 2W_3 - 3W_2^2 + 3W_4)}{1 - 4W_2 + 6W_3 - 3W_4}. \tag{55}$$